

REVIEW

Rhizosheath: An adaptive root trait to improve plant tolerance to phosphorus and water deficits?

Mehtab Muhammad Aslam^{1,2,3}  | Joseph K. Karanja¹  | Ian C. Dodd⁴ | Muhammad Waseem⁵ | Xu Weifeng^{1,2} 

¹Center for Plant Water-Use and Nutrition Regulation, College of Resource and Environment, Fujian Agriculture and Forestry University, Fuzhou, Fujian, China

²College of Agriculture, Yangzhou University, Yangzhou, Jiangsu, China

³State Key Laboratory of Agrobiotechnology, School of Life Sciences, The Chinese University of Hong Kong, Shatin, Hong Kong

⁴The Lancaster Environment Centre, Lancaster University, Lancaster, UK

⁵Department of Botany, University of Narowal, Narowal, Pakistan

Correspondence

Ian C. Dodd, The Lancaster Environment Centre, Lancaster University, Lancaster LA1 4YQ, UK.
Email: i.dodd@lancaster.ac.uk

Xu Weifeng, Center for Plant Water-Use and Nutrition Regulation, College of Resource and Environment, Fujian Agriculture and Forestry University, Fuzhou, Fujian 350002, China.
Email: wfxu@fafu.edu.cn

Funding information

National Natural Science Foundation of China; National Key R&D Program of China; Newton Advanced Fellowship

Abstract

Drought and nutrient limitations adversely affect crop yields, with below-ground traits enhancing crop production in these resource-poor environments. This review explores the interacting biological, chemical and physical factors that determine rhizosheath (soil adhering to the root system) development, and its influence on plant water uptake and phosphorus acquisition in dry soils. Identification of quantitative trait loci for rhizosheath development indicate it is genetically determined, but the microbial community also directly (polysaccharide exudation) and indirectly (altered root hair development) affect its extent. Plants with longer and denser root hairs had greater rhizosheath development and increased P uptake efficiency. Moreover, enhanced rhizosheath formation maintains contact at the root-soil interface thereby assisting water uptake from drying soil, consequently improving plant survival in droughted environments. Nevertheless, it can be difficult to determine if rhizosheath development is a cause or consequence of improved plant adaptation to dry and nutrient-depleted soils. Does rhizosheath development directly enhance plant water and phosphorus use, or do other tolerance mechanisms allow plants to invest more resources in rhizosheath development? Much more work is required on the interacting genetic, physical, biochemical and microbial mechanisms that determine rhizosheath development, to demonstrate that selection for rhizosheath development is a viable crop improvement strategy.

KEYWORDS

alternate wetting and drying cycles, drought, QTLs, water uptake

1 | INTRODUCTION

Climate change has generally increased the frequency and intensity of drought in the world's arable soils, thereby restricting crop yields (Fahad et al., 2017; Potopová et al., 2016). Crop productivity is also

influenced by its nutritional status such as phosphorus (P) deficiency, which alters plant physiological functions and limits yields (Bista et al., 2018; Farooq et al., 2021). Drought stress and P deficiency frequently co-occur since any restriction in transpiration limits the uptake of water and nutrients, and may cause agronomic effects that

Mehtab Muhammad Aslam and Joseph K. Karanja are contributed equally to this study.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2022 The Authors. *Plant, Cell & Environment* published by John Wiley & Sons Ltd.

cannot be predicted from analysis of individual stress factors (Bista et al., 2018; Suzuki et al., 2014; Xia et al., 2020; Xu et al., 2020). Drought exacerbates P deficiency in plants by reducing its uptake from the soil, transport, and redistribution in *planta* (Rouphael et al., 2012), with plant uptake restricted by diffusional limitations and/or limited root growth when fertilizers are surface-applied in water-limited environments with drying topsoils (Q. Ma et al., 2009). Moreover, P deficiency intensifies drought responses by decreasing root hydraulic conductance (Fan et al., 2007; Radin & Eidenbock, 1984; Sardans & Peñuelas, 2012). Exogenous P application enhanced maize drought tolerance by enhancing leaf gas exchange and biomass accumulation (Zahoor et al., 2021). Persistent drought during plant growth, largely driven by low precipitation and/or high evapotranspiration rates, ultimately decreases plant water and nutrient status and hence crop productivity (Gutschick & Bassirirad, 2003; Morison & Morecroft, 2006).

Phosphorus deficiency alters P assimilation at the root surface (Xia et al., 2020), root architecture, biomass allocation and inhibits shoot growth (Cho et al., 2021) and seed development, thereby limiting plant productivity. Plants have adopted sophisticated strategies to scavenge inorganic phosphate (Pi) and enhance its acquisition from P-limited soils (Ticconi et al., 2004). These include enhanced exudation of organic anions and phosphatases, activation of membrane localized P-transporters, and establishing symbiotic associations with soil microorganisms. Enhanced organic acid exudation improved P uptake of *Medicago sativa* when grown in acidic soil with high aluminium (Al) concentrations (Tesfaye et al., 2001), while *Glycine max* adapts to P deficiency by increasing root malate exudation (Liang et al., 2013). Furthermore, P solubilizing enzymes (phosphatases) might have an important role in P acquisition. For example, growth of an *Arabidopsis* mutant (*Atpap10*—for a root-secreted acid phosphatase) was substantially reduced compared to wild-type (WT) plants when grown on media containing organophosphates (L. Wang et al., 2014). Since some of these strategies incur a metabolic cost to the plants, selecting P-efficient crops is important to agriculture (Jin et al., 2006; Tariq et al., 2017; H. Zhang, Shi et al., 2020).

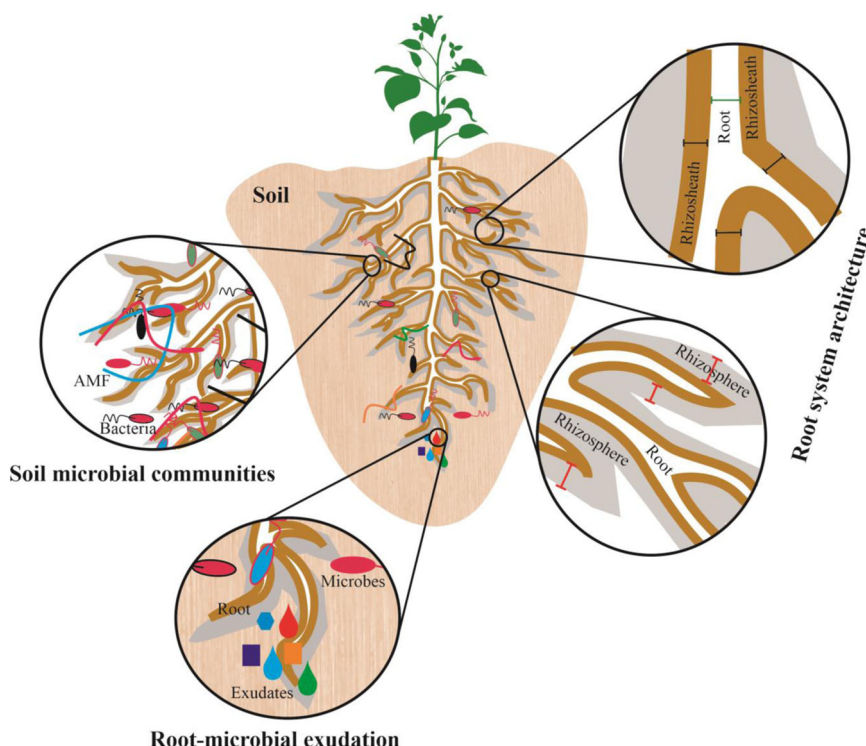
The rhizosheath is a root adaptive trait, literally defined as the weight of soil that remains attached to roots on excavation (George et al., 2014; McCully, 1999), that has been correlated with variation in plant tolerance to water deficit (Hartnett et al., 2013). A clear distinction exists between rhizosheath (the soil that adheres to the root surface on excavation), and rhizosphere, which is often defined as the volume of soil surrounding plant roots influenced by root exudates that stimulate microbial abundance (Mathesius, 2015). There is considerable genetic diversity in the ability of roots to bind soil: with some angiosperm species forming little rhizosheath (e.g., *Allium* spp; radish, *Raphanus sativus*; soybean, *G. max*), but others showing substantial rhizosheath development (e.g., *Dactylis glomerata*; alfalfa, *M. sativa*). Both monocotyledons and eudicotyledons show a similar range of specific rhizosheath weight (the ratio of bound soil mass to root mass) although in the former the order Poales (the grasses) bound more soil than order Commelinales (Brown et al.,

2017). The rhizosheath of grass species exposed to drought stress is thicker and more stable, potentially enhancing water uptake efficiency (Rabbi et al., 2018; Watt et al., 1994), and allowing plants to adapt to drought episodes (Y. Zhang, Du, Gui, et al., 2020). Although the rhizosheath has been much studied in desert grasses (Bergmann et al., 2009; Danin, 1996; Hartnett et al., 2013; Othman et al., 2004), relatively few studies have focused on the role of rhizosheath in drought resistance of crop plants (Liu, Chen et al., 2019). In dry soils, rhizosheaths have higher water content than bulk soil, and may substantially contribute to water uptake (North & Nobel, 1997). P deficiency facilitates rhizosheath formation, with barley (*Hordeum vulgare*) genotypes showing 18% larger rhizosheaths under P-deficient than P-sufficient conditions (Brown et al., 2012). Soil drying and P deficiency stimulated rhizosheath development of white lupin (*Lupinus albus*) mature cluster roots, thereby enhancing P uptake (Aslam, Karanja et al., 2021). Moreover, rhizosheath mass was positively correlated with P uptake under dry conditions (Aslam, Karanja et al., 2021; George et al., 2014), suggesting that the rhizosheath is a pivotal root trait to enhance crop nutrient status and improve drought tolerance via various mechanisms.

Rhizosheath formation involves complex interactions of multiple factors in the rhizosphere (Figure 1), the region surrounded by plant roots and whose microbial activity is highly influenced by root processes (Pang et al., 2017). These factors include root hair traits (Haling et al., 2010), plant root and microbial-derived mucilage (Carminati et al., 2017; Liu, Ye et al., 2019; Y. Zhang, Du, Xu et al., 2020), microbial activity (Hanna et al., 2013), and soil texture (Haling et al., 2014; Liu, Ye et al., 2019). Physical (root hair length and density) and chemical (root exudates) mechanisms determine rhizosheath development by enmeshing and (dis-) aggregating soil particles, respectively. Discriminating these mechanisms is difficult since root hairs can be a source of polysaccharides (Galloway et al., 2020) that bind soil particles more effectively than simpler monosaccharide molecules (Morel et al., 1991). While limited rhizosheath development of mutants (lacking root hairs) compared to WT plants demonstrates the importance of root hairs in many crop species (Burak et al., 2021), the importance of root hair traits in modifying rhizosheath development of WT plants varied. In a species (pearl millet—*Pennisetum glaucum*) with relatively short root hairs (<0.7 mm), root hair length was weakly correlated ($r^2 = 0.05$) with rhizosheath development (De la fuente Cantó et al., 2022), but in barley with longer root hairs (0.6–2.5 mm), root hair length was slightly better correlated ($r^2 = 0.16$) with rhizosheath development (George et al., 2014). Thus species vary in the importance of root hair traits in determining rhizosheath development, probably mediated by the physical dimensions of root hairs and exudate chemistry.

Furthermore, it can be difficult to dissociate plant and microbial mechanisms facilitating rhizosheath development as they can be interdependent. For example, root hairs increased axial and radial rhizosphere extension (Holz et al., 2017), which probably fosters interactions with rhizosheath-associated microbes. Root traits, most likely rhizodeposition and root hairs, are important to trap more carbon and nutrient resources in the rhizosheath, and attract

FIGURE 1 Plant-soil interactions. The rhizosheath comprises soil particles that adhere to root surface on excavation (shown as brown line). The rhizosphere is the area around plant roots influenced by root exudation and comprising its own microbial community (shown as a grey background) that is distinct from the bulk soil.



a healthy microbiome (Hallett et al., 2022). Importantly, root hair growth is finely controlled by multiple plant hormones that are modulated by drought, including abscisic acid (ABA), ethylene, auxins, and cytokinins (Gruber et al., 2013; Vissenberg et al., 2020), which thereby indirectly regulate rhizosheath formation. Moreover, microbial production and degradation of these plant hormones (Dodd et al., 2010) alters root hair development (Contesto et al., 2008), potentially affecting rhizosheath formation.

This review considers key factors that contribute to rhizosheath formation including plant genetics, the microbial community, and physiological mechanisms; then describes how they influence P status in drought-stressed environments. We also highlight specific mechanisms responsible for improving water and P availability in various crop species and finally suggest areas for further research.

2 | MECHANISMS OF RHIZOSHEATH FORMATION

2.1 | Genetics

Plants employ many molecular mechanisms to confer resistance against multiple environmental stresses of both abiotic and/or biotic origin that influence nutrient availability. Many adaptive strategies are attributed to root traits and plasticity since roots sense below-ground stresses before other plant parts (Passioura, 1988; Zheng et al., 2016), and alter their gene expression for appropriate adaptive responses (Pan et al., 2012). For instance, soil drying regulates expression of various water deprivation-responsive genes in the

roots including pathogenesis-related protein 4 (PR 4), protein phosphatase 2C (PP 2C), ABA 8'-hydroxylase, and AP2-like ER TF (Karanja et al., 2021). Nonetheless, the genetics of crucial traits affecting the rhizosheath remain elusive. Quantitative trait loci (QTLs) analysis in wheat (*Triticum aestivum*) revealed multiple loci (2B, 4D, 5A, 5B, 6A and 7A), which harboured potential basic helix-loop-helix (bHLHs) transcription factor genes associated with a robust rhizosheath. For instance, rhizosheath weight of wheat was strongly positively correlated with root hair development (Delhaize et al., 2015). bHLHs also regulate root hair length in *Oryza sativa* and *Arabidopsis thaliana* (Ding et al., 2009; Yi et al., 2010), suggesting a candidate gene underlying a rhizosheath QTL in these species. Thus, rhizosheath formation can be modulated by root hair elongation signalling via these transcription factors.

This model was reinforced by similar studies on foxtail millet (*Setaria italica*) (Liu, Ye et al., 2019), which identified QTLs associated with root hair extension, elongation, and lateral root branching located at the locus that best associated with larger rhizosheath size. In this case, five loci in *S. italica* (*Seita.3G196500*, *Seita.2G057800*, *Seita.9G333500*, *Seita.8G104600* and *Seita.7G190800*) were highly upregulated under drought stress thus hinting at their involvement in rhizosheath formation (Liu, Ye et al., 2019). Collectively, the identification of several loci in the genomes of crop species that affect rhizosheath weight suggests that multiple genes are responsible for rhizosheath formation.

Some genes such as *TaALMT1* (aluminium-activated malate transporter) may assist rhizosheath formation by secreting malate, thereby conferring Al-resistance and protecting fine and coarse root elongation allowing root development in acidic soils, while

Al-sensitive lines showed reduced rhizosheath mass and root growth in wheat (Delhaize et al., 2015). In *Sorghum bicolor*, the *SbMATE* (multidrug toxic compound extrusion) involved in root growth processes (Carvalho et al., 2016) secretes organic acid (citrate), which has been hypothesized as a potent stimulator of soil microbial activity (Macias-Benitez et al., 2020). However, a recent study identified 12 potential QTLs may control rhizosheath formation in pearl millet, indicating complex genetic regulation mainly via root exudation (De la fuente Cantó et al., 2022). Therefore, these genes/QTLs associated with root growth and stress-responsiveness may improve abiotic stress tolerance. Taken together, these findings suggest various genes (Table 1) that are functionally significant in rhizosheath formation via mechanisms attributed to root growth and organic acid exudation, but their effects vary between species and rhizosheath formation is a multi-genic phenomenon.

2.2 | Physico-chemical factors

Drying and wetting cycles promote soil aggregation by modifying the soil pore system, with soil pores being filled with water during wetting, and irreversible rearrangement of soil particles during drying (Pires et al., 2007). This process interacts with other gluing agents released in the root-zone by the roots themselves or soil microbes, playing complementary roles in promoting soil particle aggregation and adhesion to the root surface (rhizosheath formation). Drier soils enhanced adhesiveness of the rhizosheath and facilitated root hair formation; both of which are involved in rhizosheath stabilisation (Watt et al., 1994).

Soil characteristics such as texture, aggregate stability, organic matter content, number of drying and rewetting cycles, and soil water content all affect rhizosheath stability (Ghezzehei & Albalasmeh, 2015; Haling et al., 2010; Haling et al., 2014). Roots

growing in soils with higher (80%) sand content had more root hairs and more pronounced rhizosheaths than those growing in lower (30%) sand content (Bailey & Scholes, 1997). Barley genotypes (differing in root hair development) had greater rhizosheath development when grown in a sandy loam than a clay loam (Marin et al., 2021). When grown in a sandy loam soil compacted to a range of bulk densities ($1.2\text{--}1.7\text{ g cm}^{-3}$), barley rhizosheath development was greatest at the lowest bulk density (1.2 g cm^{-3}) but declined at next highest (1.4 g cm^{-3}) despite similar root hair lengths, indicating the importance of both soil physical and chemical processes (Haling et al., 2014). Fine textured soils were predicted to produce more water stable aggregates at a lower soil organic matter content when exposed to drying and rewetting cycles (Albalasmeh & Ghezzehei, 2014). Thus physical and chemical processes interact in rhizosheath formation.

Rhizosheath formation alters soil particle size distribution around the roots, increasing the frequency of smaller particles compared to the bulk soil (Wei et al., 2011). Water fluxes associated with plant transpiration move colloidal particles within the soil towards the roots (Albalasmeh & Ghezzehei, 2014), while disaggregation of soil particles in response to the exudation of organic acids also contributes (Naveed et al., 2017). The size of soil pores relative to root hair diameter likely determines the importance of physical versus chemical processes in determining root adhesion to the surrounding soil. In dense soils, roots encounter series of channels, and biopores, or grow between regions of soil of contrasting strength. More than 85% of wheat roots deep in the soil profile (below 50 cm) were clumped within biopores and channels (White & Kirkegaard, 2010), with substantial root hair proliferation along the pore wall acting to maintain hydraulic root-soil contact. Moreover, root hairs may contribute to anchorage to assist the penetration of individual root tips and increase the soil-root binding (De Baets et al., 2020).

TABLE 1 Genes/QTLs related to rhizosheath formation in different crop species

Plant	Gene/QTLs	Role in rhizosheath formation	References
Transgenic barley	Overexpressed with wheat gene <i>TaALMT1</i>	Enhanced malate efflux, efficient P uptake and grain production.	Delhaize et al. (2009)
Barley	QTL on chromosome 2H	Rhizosheath weight correlated with P uptake under dry conditions	George et al. (2014)
Pearl millet	Twelve QTLs	Rhizosheath formation was complex, mainly regulated by root exudation	De la fuente Cantó et al. (2022)
Foxtail millets	<i>Seita.3G196500</i> , <i>Seita.9G333500</i> , <i>Seita.8G104600</i> , <i>Seita.2G057800</i> and <i>Seita.7G190800</i>	Showed higher transcript level under drought conditions	Liu, Ye et al. (2019)
Rice	<i>L. albABCG29</i>	Improved P use through enhanced root growth and increased rhizosheath formation under low P soil drying	Aslam, Wassem et al. (2021)
Wheat	Chromosome no 2B, 4D, 5A, 5B, 6A and 7A and five major loci	Contribute to 42% rhizosheath variations, accounting for over 60% of the total genetic variance	Delhaize et al. (2015); James et al. (2016); Marr (2021)

Abbreviation: Quantitative trait loci.

Alternate wetting and drying cycles (driven by evapotranspiration) interact with mucilage (a viscoelastic gel substance rich in polysaccharides) at the root-soil interface to stimulate soil structure changes and increase rhizosheath porosity, as determined using microcomputed tomography (Y. Zhang, Du, Gui, et al., 2020). The quantity and quality of polysaccharides, and the ambient soil matrix potential both play an important role in the mucilage's ability to aggregate soil. Mucilage greatly affects soil water dynamics. During soil drying, it sustains higher soil water content which potentially enhances soil aggregation, but at lower water potentials mucilage becomes hydrophobic and decreases rhizosphere hydraulic conductivity (Kroener et al., 2014; Rosenzweig et al., 2009). With prolonged soil drying, mucilage is suggested to lose water to the surrounding soil (Ahmed et al., 2014), thereby increasing its viscosity and in turn decreasing its surface tension. Decreased surface tension increases their ability to wet the surrounding soil (Read & Gregory, 1997), while high viscosity of exudates increases the resistance of adhering soil particles to movement (Moradi et al., 2012; Read & Gregory, 1997). Consequently, rhizosheath stability and thickness increases (Walker et al., 2003). Interestingly, the rhizosheath of drought-tolerant grasses (growing in the African savannah) was significantly ($p < 0.0001$) thicker (by 65%) than in drought-sensitive grasses (Hartnett et al., 2013), suggesting that the rhizosheath enhances rhizosphere water retention capacity under drought stress, thereby promoting plant growth and survival.

Exudates/mucilage enable plants to create microenvironments which can be beneficial for plant growth. The release of polysaccharide-rich mucilage from the root tips eases root penetration in deeper soil layers and may serve as a protective barrier (Galloway et al., 2020). Since microbial and root contributions to rhizosheath development can be difficult to distinguish, future experiments should aim to exclude the confounding effects of microbial community-derived mucilage secretions. Plants could be grown in clean river sand as it has low microbial activity (Marasco et al., 2018; Neilson et al., 2017) or soils could be sterilized by gamma irradiation or other ways (Cheptsov et al., 2021; Mahmood et al., 2014; Yin & Wang, 2016). Such practices may better distinguish the relative contribution of root versus microbial mucilage to rhizosheath formation.

2.3 | Microbial community

Severe drought stress decreases the size and activity of rhizosphere microbial biomass (Sanaullah et al., 2011). Microbes have adopted various strategies to thrive in the rhizosphere niche (Aslam et al., 2022; Jacoby et al., 2017). Many bacterial genera release exopolysaccharides (EPS) (Costa et al., 2018) including *Bacillus*, *Pseudomonas* and *Azospirillum brasilense*, which permeate the surrounding soils bonding them together and increasing their aggregation to the root surface as mucilage dries (Walker et al., 2003). Soil drying stimulates EPS production, thereby enhancing soil moisture retention and buffering bacterial colonies from environmental stress (Roberson &

Firestone, 1992). Soil inoculation with EPS-producing bacteria, that produce more EPS as the soil dries, can enhance soil aggregation and rhizosheath development, thereby improving leaf water status when plants are grown in drying soil (Alami et al., 2000; Sandhya et al., 2009). Thus, it has been postulated that rhizosheath-associated microbial communities play a profound role in forming a coherent rhizosheath (McCully & Canny, 1988; Watt et al., 1993).

Certain classes of soil microbes released volatile organic compounds (including esters, ethers, aldehydes, naphthyl derivatives, ketones, alkyls and benzene derivatives) which facilitate microbial interactions (Yuan et al., 2017) and induce root secretion (Phillips, 2007; Phillips et al., 2008) or modification (Fernandez et al., 2016) of plant metabolites such as organic acids, which greatly influence rhizosheath formation. Bacterial exudation of various compounds can alter rhizosheath water holding capacity during soil drying. Wheat plants inoculated with *Bacillus* (T-34) and *Azospirillum* (WS-1) strains promoted root growth and increased rhizosheath dry weight compared to non-inoculated plants under both wheat-cotton and wheat-rice crop rotations (Tahir et al., 2015). Inoculating EPS-producing bacteria could maintain soil water content to alleviate drought stress (Ashraf et al., 2004), suggesting these beneficial bacteria could serve as a promising strategy to alleviate abiotic stresses. Optimal water content around the root is associated with increased enzymatic activity, typically phosphatase activity (Guenet et al., 2012), which assists phosphate acquisition in plants, consequently promoting crop growth (Hu et al., 2013; Turan et al., 2012). However, it can be difficult to determine the role of specific microbial traits in enhancing rhizosheath development and plant stress tolerance, as many bacteria have multiple beneficial traits. For example, EPS-producing bacteria can also produce ammonia, hydrogen cyanide and the phytohormone indole 3-acetic acid (Khan & Bano, 2019), all of which might affect root and rhizosheath development in drying soil. More specific evidence on the importance of microbial EPS in rhizosheath development should be sought by obtaining bacterial mutants that lack EPS production (Deka et al., 2019) and determining their impact when plants are grown in drying soil.

A recent example utilized bacterial mutagenesis to demonstrate the impacts of particular microbial traits on rhizosheath development (Y. Zhang, Du, Xu et al., 2020). Inoculating rice with the aminocyclopropane-1-carboxylic acid (ACC) deaminase containing rhizobacterium *Enterobacter aerogenes* G3 almost doubled rhizosheath formation compared to its ACC-deaminase minus mutant, presumably since the WT strain produced significantly longer root hairs, physically enmeshing the soil. Surprisingly, previous in vitro investigations with a range of WT rhizobacteria and their ACC-deaminase minus mutants indicated that the latter produced significantly longer root hairs (Contesto et al., 2008), although the effects of perturbing root ethylene status on root hair development will depend on other microbially produced hormones and the root growth environment. Alternatively, adding the ACC deaminase gene to *A. brasilense* increased root hair density of *Vicia sativa* (Star et al., 2012), which should promote rhizosheath development. Interestingly,

the rice rhizosheath was enriched with *Enterobacteriaceae* (Y. Zhang, Du, Xu et al., 2020), leading to the hypothesis that crop species may select specific taxa of beneficial soil microbes to improve water status through unidentified mechanisms, presumably via signals derived from root exudates. Whether there are genetic determinants that underpin how different plant species interact with rhizospheric microbial communities needs attention, as such discoveries may provide potential targets for breeding water efficient crops.

3 | RHIZOSHEATH INCREASES PHOSPHORUS UPTAKE BY PLANT ROOTS

Phosphorus is an essential macronutrient for crop growth and yield, yet it is poorly available in the soil for plant uptake owing to its low diffusion rate (Lambers et al., 2015, 2018). To cope with P deficiency and support growth-related processes, plants forage the soil for this sparingly available resource, by enhancing lateral root branching and root hair development (Lambers et al., 2015; Lynch & Brown, 2001; Lynch, 2019). Long root hairs increase the surface area for intercepting P diffusing towards the roots (Haling et al., 2016). This root-hair-cylinder, in most cases, positively correlates with rhizosheath size (Haling et al., 2010). Enhanced rhizosheath development alleviated P deficiency stress of barley and wheat (Brown et al., 2012; Haling et al., 2013). Hence, while rhizosheath formation and root hair growth are correlated with enhanced

P uptake efficiency, other mechanisms by which rhizosheath facilitates uptake of P also exist.

In P-efficient genotypes growing in low P availability, plant roots release exudates to mobilize nutrients in the rhizosphere (Krishnapriya & Pandey, 2016; Nazari, 2021). Components of these root exudates help plants to access nutrients by acidifying the rhizosphere or chelating nutrients (Raghothama, 1999). Oat (*Avena sativa*) and sugar beet (*Beta vulgaris*) plants were grown in roots split between soil compartments, to expose part of the root system to dry soil while keeping the other soil compartment moist to avoid leaf water deficit. Although decreased K inflow was consistent with a modelled diffusional impediment, soil moisture content did not influence P inflow due to increased root exudation rate, especially of high molecular weight compounds from water-stressed plants, which increased P concentrations in the soil solution (Lieberbach et al., 2004). Thus, exudation of mucilage could be a promising route to overcome P uptake and transport under soil drying events (Figure 2).

Malate, citrate, and malonate are potent organic acids exuded by various crop species. Carboxylate (malonate and citrate) secretion by legumes and white lupin solubilizes soil P, making it accessible for roots to capture (Dinkelaker et al., 1989; Hocking et al., 1997; Kidd et al., 2016). The lupin rhizosheath had a higher carboxylate content than those of pasture legumes such as *Trifolium* and *Medicago* species (Kidd et al., 2016), suggesting that lupin may better mobilize minimally available P from the rhizosphere. Thus, P homeostasis

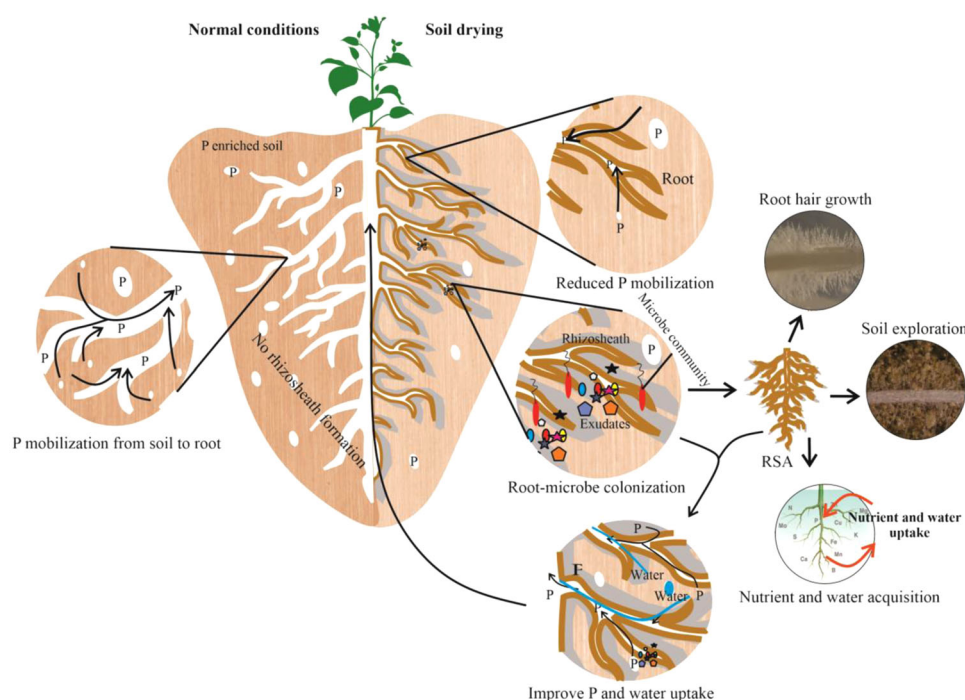


FIGURE 2 Soil drying and its effects on plant tissue P concentrations. Soil drying events activate root exudation which in turn enhances rhizosheath formation. Root exudates mediate P-deprivation signalling, ultimately altering RSA which improves soil exploration for nutrient and water uptake. Additionally, root exudation initiates root microbial colonisation in the rhizosphere. Together, organic acids exuded by plant roots and microbes promote phosphorus solubilisation. RSA, Root system architecture; P, phosphorus. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pcpe.14395)]

likely involves complex and highly coordinated processes with contributions potentially from root system development and physico-chemical components of root exudates and microbial secreted mucilage (Figure 2). Cover crop (*Sinapis alba*, *Phacelia tanacetifolia*, and *Fagopyrum esculentum*) rhizosheath soil had higher microbial abundance than the bulk soil (Hallama et al., 2022), which was partially attributed to rhizodeposition (presumably organic acids) providing a C-source for microbes associated with P cycling (M. S. Khan et al., 2014) which can enhance availability of P for plant use (Richardson & Simpson, 2011). This is likely since plant roots exude organic acids and extracellular enzymes (phytase and acid phosphatase) to mobilise inorganic P from Al, Ca or Fe phosphates (Felix & Donald, 2002), or organic P compounds (Chen et al., 2006; J. F. Ma et al., 2001), allowing absorption by the plant. Upon P deficiency, plants release specific root exudates like gamma-aminobutyric acid (GABA, acting as a chemo-attractant for P-solubilizing microbes), and acidify the rhizosphere to make additional Pi available (Dakora & Phillips, 2002; Felix & Donald, 2002; Hinsinger, 2001). These processes ultimately improve nutrient acquisition and availability in nutrient-depleted soils.

Rhizosphere bacteria can independently (Jeong et al., 2013; Qureshi et al., 2012; Scagliola et al., 2016) or via interaction with plants (Pii et al., 2015; Segura & Ramos, 2013), secrete P-solubilizing enzymes and organic acids, thus promoting absorption of previously insoluble nutrients by plant (Pii et al., 2015, 2016; Rodriguez et al., 2004). However, how root exudates reprogramme the root microbiome (Pascale et al., 2020) is still not clear. Cover crops can increase soil P availability by enhancing the abundance and activity of P-solubilising microbes in the rhizosheath (Hallama et al., 2022). Although integrative analyses of microbial composition and microbial functional genes associated with P cycling remain significantly challenging owing to variability of P in the field, they may play a crucial role in P acquisition in the rhizosheath. Similarly, high rhizosheath phosphatase activity has been associated with enhanced P acquisition, thereby increasing root and shoot biomass, and P accumulation (Hunter et al., 2014). Since rhizodeposition by the root system nourishes the rhizobiome and promotes rhizosheath formation, further studies determining expression of rhizosheath phosphatase activity genes would extend our understanding of rhizosheath-dependent P acquisition.

4 | RHIZOSHEATH IMPACTS ON PLANT WATER UPTAKE

Empirical evidence demonstrates that soil moisture in the rhizosheath was substantially higher than that of the surrounding bulk soil in different plant species, according to the absolute soil water potential (Moradi et al., 2012; Young, 1995). While wheat and barley roots had hydrophilic mucilage (Naveed et al., 2017; Zickenrott et al., 2016), some rhizosheaths were more hydrophobic than others (Brown et al., 2017), with maize, lupins and chia roots having hydrophobic mucilage (Zeppenfeld et al., 2017). Furthermore, a modelling exercise (Carminati et al., 2011) showed that mucilage attenuates gradients

in water potential in the rhizosphere, thus increasing water acquisition by roots as the soil dries.

Rhizosheath formation can act as a 'bridge' to the bulk soil, thereby minimising the formation of air gaps adjacent to the root system, which substantially increase hydraulic resistance at the soil-plant interface (North & Nobel, 1997). Mucilage exuded by the roots limits the development of air gaps around roots, and improves the ability of plant roots to take up water by decreasing hydraulic resistance in the rhizosphere (Ahmed et al., 2014; Carminati et al., 2014). Nevertheless, analysing root shrinkage (air gap formation) in relation to soil water status as broad bean (*Vicia faba*) and white lupin plants dried a sandy soil (Carminati & Vetterlein, 2013; Koebernick et al., 2018) demonstrated that root shrinkage occurred after transpiration rate declined, suggesting that other factors than root-soil contact were limiting water uptake.

While air gaps may restrict root water uptake from the soil, they may also reduce water loss from roots towards the soil (Carminati & Vetterlein, 2013; Zarebanadkouki & Carminati, 2014) which becomes important when soil water potential is lower than root water potential. Although mucilage secretion may initially hydrate the rhizosheath, it may later limit water loss as soil drying progresses (Ahmed et al., 2014; Carminati & Vetterlein, 2013). The gelling effect of mucilage within the rhizosheath can decrease water flow from the rhizosheath soil to the bulk soil distant from the root surface, presumably by reducing the matric potential gradient between rhizosheath and the bulk soil (Ahmed et al., 2016; Carminati et al., 2016). Prolonged soil drying can make the mucilage water-repellent (Czarnes et al., 2000; Lichner et al., 2002; Moradi et al., 2012), thus limiting plant water uptake when the soil is re-watered, and delaying the recovery of leaf water potential and stomatal conductance. Thus the adaptive value of mucilage will vary according to the severity and frequency of drying episodes, and its chemical composition.

Several studies have compared the responses of mutants lacking root hairs with their respective WT, although effects of root hairs *per se*, rather than their contribution to rhizosheath formation, cannot be distinguished. When the bald root barley (*brb*) mutant and its WT were exposed to abrupt changes in evaporative demand, the mutant could not sustain transpiration rates at high vapour pressure deficits (VPDs) without substantially decreasing leaf water potential (Carminati et al., 2017). At more moderate VPDs (<1.5 kPa), there were no genotypic differences in transpiration rate, leaf water potential or xylem ABA concentration (Dodd & Diatloff, 2016), indicating that root hairs (and presumably rhizosheath formation) were redundant in sustaining water uptake. Alternatively, wheat cultivars with larger rhizosheaths sustained higher transpiration rates as the soil dried, while contrasting cultivars with diminished rhizosheath wilted more readily (Basirat et al., 2019). Further work is required using imaging techniques (X-ray Computer Tomography and magnetic resonance imaging) to determine the timing of air gap formation relative to the decline in transpiration in genotypes that differ in rhizosheath development. To be confident of rhizosheath impacts on plant water relations, future work should aim to manipulate rhizosheath formation via different mechanisms (not just

variation in root hair development) to determine whether physiological responses to variation in rhizosheath development are consistent.

5 | ALTERNATE WETTING AND DRYING OF RICE TO ENHANCE WATER AND P USE EFFICIENCY

Alternate drying and rewetting cycles (ADW) is a deliberate strategy used to save water in agriculture, thereby improving crop water use efficiency (Dodd et al., 2015). Repeated ADW cycles in rice paddy fields dynamically alters the microbial community, soil water and nutrient availability (Watanabe et al., 2021). Allowing rice fields to periodically dry out by withholding irrigation can decrease water use by 20%–40% compared to continuous flooding, without influencing rice production or yield (Carrijo et al., 2017; Rejesus et al., 2011). While part of this water saving will occur through physical processes (decreased evaporation and drainage/leakage from rice paddies), stomatal closure can also occur if soil matric potential declines sufficiently (Li et al., 2017). Such stomatal closure cannot always be attributed to decreased leaf water potential (Dingkuhn et al., 1989; X. Wang et al., 2018), but instead to the root-to-shoot transport of chemical signals like ABA (Bano et al., 1993; Siopongco et al., 2008). Irrespective of the mechanisms of stomatal closure, intrinsic water use efficiency increases (Caine et al., 2019), since transpiration is restricted more than photosynthesis (Hidayati & Anas, 2016). Selecting genotypes with variation in the production of, or sensitivity to, root-to-shoot signals (especially ABA) under progressive soil ADW or drought may enhance crop water use efficiency and sustain yields under environmental stresses, although it is less clear that these signalling processes are related to rhizosheath development.

Multiple mechanisms inhibit plant growth as the soil dries, including decreased P uptake and transport (Rouphael et al., 2012). Upon rewetting, physico-chemical processes that disrupt dried soil aggregates release P into the soil solution (Bünemann et al., 2013), along with lysis of microbial cells associated with osmotic shock (Blackwell et al., 2010; Bünemann et al., 2013; Turner et al., 2003). Thus, the impacts of ADW cycles (especially in periodically flooded soils such as in rice cultivation) on plant P uptake can be difficult to predict, because these different microbial and physico-chemical processes affect soil P availability. Repeated ADW events confer resistance to some microbial communities rendering them less susceptible to osmotic stress-induced lysis (Butterly et al., 2009; Sawada et al., 2016). The frequency of ADW can influence the microbial community composition, with microbes capable of recovering from desiccation sustaining the microbial community (Zhao et al., 2016) and solubilizing P allowing plant uptake. Both ADW events (Deneff et al., 2001), and root mucilage (Chaparro et al., 2014; Haichar et al., 2012; Kawasaki et al., 2016; Terrazas et al., 2016) can select for specific microbial communities that are tolerant to rewetting stresses or communities that are specific to plant species, respectively. Possibly rhizosheath formation sustains these microbial communities, helping plants to cope with nutrient and water constraints.

Whether ADW contributes to rhizosheath formation to enhance P use efficiency of rice is still uncertain, especially since it may be difficult to measure rhizodeposition (Kuzyakov & Domanski, 2000) and nutrient concentrations in the soil solution (Norton et al., 2017) in such systems. Since root exudates involved in rhizosheath formation are a crucial adaptive strategy for plant P acquisition from P-deficient soils (Ndour et al., 2020), such findings imply that ADW may potentially contribute to P uptake via rhizosheath formation. Experimental evidence of precise mechanisms underlying the complementary roles of ADW and rhizosheath formation in determining water and P use efficiency is still scarce.

6 | CONCLUSIONS AND FUTURE PERSPECTIVES

Rhizosheath formation requires complex interactions of root hair traits with root- and microbe-derived mucilage. To assist plant breeding efforts, it is imperative to uncover the underlying genetic mechanism(s) that promote efficient water and phosphorus use under suboptimal soil conditions, and identifying genes regulating rhizosheath formation may be important in this endeavour. While most work demonstrating the potential importance of rhizosheath development in regulating plant physiological responses to soil conditions has compared WT plants and mutants lacking root hairs, exploiting genetic diversity in root hair traits and root exudation profiles within diverse germplasm of major crop species seems important. Since these inter-connected physical and chemical plant-mediated processes interact with multiple soil properties in determining rhizosheath formation, pronounced genotype × environment interactions are highly likely. Whether plant breeders developing crop genotypes adapted to water- and nutrient-limited environments have unconsciously selected for enhanced rhizosheath formation in specific environments remains to be determined, but enhancing these processes may be critical to further progress in improving crop resilience to these stresses.

The development of new techniques to discriminate genetic variation in processes affecting rhizosheath development will complement traditional rhizosheath measurements that excavate the root system and quantify the adhering soil. The ability of root exudates to bind soil *ex vivo* can be determined by collecting them from hydroponically grown plants, concentrating them and then applying them to nitrocellulose. Dried, sieved soil is then placed on the membrane and allowed to adhere, with unbound soil removed before quantifying that remaining on the membrane (Akhtar et al., 2018). This technique discriminated variation in exudate adhesiveness between WT and root hairless mutants (Burak et al., 2021), but has yet to be applied to large populations of genetically related plants such as recombinant inbred lines to identify potential QTLs mediating rhizosheath development. This soil-based technique complements high-throughput antibody-based tests such as ELISA (Galloway et al., 2020) that can measure root polysaccharide release in the same hydroponic samples. However, the impact of the soil environment

(moisture and P status) on root exudation requires further investigation, to ensure that these ex vivo assays correlate with in vivo rhizosheath development.

While it is clearly evident that stimulation of rhizosheath development in drying soil is dependent on root hair traits and root exudation, further work focusing on the association between root exudate chemistry and expression profiles of P-uptake genes is warranted. Moreover, further exploration of the association between rhizosheath traits (weight, strength and porosity) and P uptake are necessary in crop species grown under varied soil types and contrasting water levels. Such studies would help define the utility of rhizosheath in crop breeding for greater sustainability.

ACKNOWLEDGEMENTS

This study was supported by grants from the National Key R&D Programme of China (2017YFE0118100, 2018YFD02003025 and 2018YFD1000800), National Natural Science Foundation of China (Nos. 31872169, 31722048, and 31761130073) and a Newton Advanced Fellowship (NSFC-RS: NA160430).

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ORCID

Mehtab Muhammad Aslam  <https://orcid.org/0000-0002-0890-4819>

Joseph K. Karanja  <https://orcid.org/0000-0003-2338-0926>

Xu Weifeng  <https://orcid.org/0000-0002-9424-991X>

REFERENCES

- Ahmed, M.A., Kroener, E., Benard, P., Zarebanadkouki, M., Kaestner, A. & Carminati, A. (2016) Drying of mucilage causes water repellency in the rhizosphere of maize: measurements and modelling. *Plant and Soil*, 407, 161–171.
- Ahmed, M.A., Kroener, E., Holz, M., Zarebanadkouki, M. & Carminati, A. (2014) Mucilage exudation facilitates root water uptake in dry soils. *Functional Plant Biology*, 41, 1129–1137.
- Akhtar, J., Galloway, A.F., Nikolopoulos, G., Field, K.J. & Knox, P. (2018) A quantitative method for the high throughput screening for the soil adhesion properties of plant and microbial polysaccharides and exudates. *Plant and Soil*, 428, 57–65.
- Alami, Y., Achouak, W., Marol, C. & Heulin, T. (2000) Rhizosphere soil aggregation and plant growth promotion of sunflowers by an exopolysaccharide-producing *Rhizobium* sp. strain isolated from sunflower roots. *Applied and Environmental Microbiology*, 66, 3393–3398.
- Albalasmeh, A.A. & Ghezzehei, T.A. (2014) Interplay between soil drying and root exudation in rhizosheath development. *Plant and Soil*, 374, 739–751.
- Ashraf, M., Hasnain, S., Berge, O. & Mahmood, T. (2004) Inoculating wheat seedlings with exopolysaccharide-producing bacteria restricts sodium uptake and stimulates plant growth under salt stress. *Biology and Fertility of Soils*, 40, 157–162.
- Aslam, M.M., Idris, A.L., Zhang, Q., Weifeng, X.U., Karanja, J.K. & Wei, Y. (2022) Rhizosphere microbiomes can regulate plant drought tolerance. *Pedosphere*, 32, 61–74.
- Aslam, M.M., Karanja, J.K., Yuan, W., Zhang, Q., Zhang, J. & Xu, W. (2021a) Phosphorus uptake is associated with the rhizosheath formation of mature cluster roots in White lupin under soil drying and phosphorus deficiency. *Plant Physiology and Biochemistry*, 166, 531–539.
- Aslam, M.M., Waseem, M., Zhang, Q., Ke, W., Zhang, J. & Xu, W. (2021b) Identification of ABC transporter G subfamily in White lupin and functional characterization of *L. alba*ABGC29 in phosphorus use. *BMC Genomics*, 22, 1–14.
- Baets, D.E., Denbigh, S., Smyth, T.D.G., Eldridge, K.M., Weldon, B.M., Higgins, L. et al. (2020) Micro-scale interactions between Arabidopsis root hairs and soil particles influence soil erosion. *Communications Biology*, 3, 164.
- Bailey, C. & Scholes, M. (1997) Rhizosheath occurrence in South African grasses. *South African Journal of Botany*, 63, 484–490.
- Bano, A., Dorffling, K., Bettin, D. & Hahn, H. (1993) Absciscic acid and cytokinins as possible root-to-shoot signals in xylem sap of rice plants in drying soil. *Functional Plant Biology*, 20, 109–115.
- Basirat, M., Mousavi, S.M., Abbaszadeh, S., Ebrahimi, M. & Zarebanadkouki, M. (2019) The rhizosheath: a potential root trait helping plants to tolerate drought stress. *Plant and Soil*, 445, 565–575.
- Bergmann, D., Zehfus, M., Zierer, L., Smith, B. & Gabel, M. (2009) Grass rhizosheaths: associated bacterial communities and potential for nitrogen fixation. *Western North American Naturalist*, 69, 105–114.
- Bista, D., Heckathorn, S., Jayawardena, D., Mishra, S. & Boldt, J. (2018) Effects of drought on nutrient uptake and the levels of nutrient-uptake proteins in roots of drought-sensitive and-tolerant grasses. *Plants*, 7, 28.
- Blackwell, M., Brookes, P., La Fuente-Martinez, D.E., Gordon, N., Murray, H., Snars, P. et al. (2010) Phosphorus solubilization and potential transfer to surface waters from the soil microbial biomass following drying–rewetting and freezing–thawing. *Advances in Agronomy*, 106, 1–35.
- Brown, L.K., George, T., Thompson, J., Wright, G., Lyon, J., Dupuy, L. et al. (2012) What are the implications of variation in root hair length on tolerance to phosphorus deficiency in combination with water stress in barley (*Hordeum vulgare*)? *Annals of Botany*, 110, 319–328.
- Brown, L.K., George, T.S., Neugebauer, K. & White, P.J. (2017) The rhizosheath—a potential trait for future agricultural sustainability occurs in orders throughout the angiosperms. *Plant and Soil*, 418, 115–128.
- Bünemann, E., Keller, B., Hoop, D., Jud, K., Boivin, P. & Frossard, E. (2013) Increased availability of phosphorus after drying and rewetting of a grassland soil: processes and plant use. *Plant and Soil*, 370, 511–526.
- Burak, E., Quinton, J.N. & Dodd, I.C. (2021) Root hairs are the most important root trait for rhizosheath formation of barley (*Hordeum vulgare*), maize (*Zea mays*) and *Lotus japonicus* (Gifu). *Annals of Botany*, 128, 45–57.
- Butterly, C., Bünemann, E., McNeill, A., Baldock, J. & Marschner, P. (2009) Carbon pulses but not phosphorus pulses are related to decreases in microbial biomass during repeated drying and rewetting of soils. *Soil Biology and Biochemistry*, 41, 1406–1416.
- Caine, R.S., Yin, X., Sloan, J., Harrison, E.L., Mohammed, U., Fulton, T. et al. (2019) Rice with reduced stomatal density conserves water and has improved drought tolerance under future climate conditions. *New Phytologist*, 221, 371–384.
- Carminati, A., Benard, P., Ahmed, M.A. & Zarebanadkouki, M. (2017) Liquid bridges at the root-soil interface. *Plant and Soil*, 417, 1–15.
- Carminati, A., Kroener, E., Ahmed, M.A., Zarebanadkouki, M., Holz, M. & Ghezzehei, T. (2016) Water for carbon, carbon for water. *Vadose Zone Journal*, 15, 1–10.
- Carminati, A., Kroener, E., Holz, M., Zarebanadkouki, M. & Ahmed, M.A. (2014) Mucilage exudation facilitates root water uptake in dry soils. *Functional Plant Biology*, 41, 1129–1137.
- Carminati, A., Schneider, C.L., Moradi, A.B., Zarebanadkouki, M., Vetterlein, D., Vogel, H.-J.R. et al. (2011) How the rhizosphere

- May favor water availability to roots. *Vadose Zone Journal*, 10, 988–998.
- Carminati, A. & Vetterlein, D. (2013) Plasticity of rhizosphere hydraulic properties as a key for efficient utilization of scarce resources. *Annals of Botany*, 112, 277–290.
- Carrijo, D.R., Lundy, M.E. & Linqvist, B.A. (2017) Rice yields and water use under alternate wetting and drying irrigation: a meta-analysis. *Field Crops Research*, 203, 173–180.
- Carvalho, G., Schaffert, R.E., Malosetti, M., Viana, J.H.M., Menezes, C.B., Silva, L.A. et al. (2016) Back to acid soil fields: the citrate transporter SbMATE is a major asset for sustainable grain yield for sorghum cultivated on acid soils. *G3: Genes, Genomes, Genetics*, 6, 475–484.
- Chaparro, J.M., Badri, D.V. & Vivanco, J.M. (2014) Rhizosphere microbiome assemblage is affected by plant development. *The ISME journal*, 8, 790–803.
- Chen, Y.P., Rekha, P.D., Arun, A.B., Shen, F.T., Lai, W.A. & Young, C.C. (2006) Phosphate solubilizing bacteria from subtropical soil and their tricalcium phosphate solubilizing abilities. *Applied Soil Ecology*, 34, 33–41.
- Cheptsov, V.S., Vorobyova, E.A., Manucharova, N.A., Gorlenko, M.V., Pavlov, A.K., Rozanova, M.S. et al. (2021) Prokaryotic community of the ancient antarctic permafrost after irradiation with gamma rays under simulated martian conditions. *Eurasian Soil Science*, 54, 417–423.
- Cho, H., Bouain, N., Zheng, L. & Rouached, H. (2021) Plant resilience to phosphate limitation: current knowledge and future challenges. *Critical Reviews in Biotechnology*, 41, 63–71.
- Contesto, C., Desbrosses, G., Lefoulon, C., Béna, G., Borel, F., Galland, M. et al. (2008) Effects of rhizobacterial ACC deaminase activity on *Arabidopsis* indicate that ethylene mediates local root responses to plant growth-promoting rhizobacteria. *Plant Science*, 175, 178–189.
- Costa, O.Y.A., Raaijmakers, J.M. & Kuramae, E.E. (2018) Microbial extracellular polymeric substances: ecological function and impact on soil aggregation. *Frontiers in Microbiology*, 9, 1636.
- Czarnes, S., Hallett, P.D., Bengough, A.G. & Young, I.M. (2000) Root- and microbial-derived mucilages affect soil structure and water transport. *European Journal of Soil Science*, 51, 435–443.
- Dakora, F.D. & Phillips, D.A. (2002) Root exudates as mediators of mineral acquisition in low-nutrient environments. *Food security in nutrient-stressed environments: exploiting plants' genetic capabilities*. pp. 201–213.
- Danin, A. (1996) Adaptations of stipagrostis species to desert dunes. *Journal of Arid Environments*, 34, 297–311.
- Deka, P., Goswami, G., Das, P., Gautam, T., Chowdhury, N., Boro, R.C. et al. (2019) Bacterial exopolysaccharide promotes acid tolerance in *Bacillus amyloliquefaciens* and improves soil aggregation. *Molecular Biology Reports*, 46, 1079–1091.
- De la fuente Cantó, C., Diouf, M.N., Ndour, P.M.S., Debieu, M., Grondin, A., Passot, S. et al. (2022) Genetic control of rhizosheath formation in pearl millet. *Scientific Reports*, 12, 9205.
- Delhaize, E., Rathjen, T.M. & Cavanagh, C.R. (2015) The genetics of rhizosheath size in a multiparent mapping population of wheat. *Journal of Experimental Botany*, 66, 4527–4536.
- Delhaize, E., Taylor, P., Hocking, P.J., Simpson, R.J., Ryan, P.R. & Richardson, A.E. (2009) Transgenic barley (*Hordeum vulgare* L.) expressing the wheat aluminium resistance gene (*TaALMT1*) shows enhanced phosphorus nutrition and grain production when grown on an acid soil. *Plant Biotechnology Journal*, 7, 391–400.
- Denef, K., Six, J., Bossuyt, H., Frey, S.D., Elliott, E.T., Merckx, R. et al. (2001) Influence of dry-wet cycles on the interrelationship between aggregate, particulate organic matter, and microbial community dynamics. *Soil Biology and Biochemistry*, 33, 1599–1611.
- Ding, W., Yu, Z., Tong, Y., Huang, W., Chen, H. & Wu, P. (2009) A transcription factor with a bHLH domain regulates root hair development in rice. *Cell Research*, 19, 1309–11.
- Dingkuhn, M., Cruz, R.T. & O'toole, J.C. (1989) Net photosynthesis, water use efficiency, leaf water potential and leaf rolling as affected by water deficit in tropical upland rice. *Australian Journal of Agricultural Research*, 40, 1171–1181.
- Dinkelaker, B., Römhild, V. & Marschner, H. (1989) Citric acid excretion and precipitation of calcium citrate in the rhizosphere of White lupin (*Lupinus albus* L.). *Plant, Cell & Environment*, 12, 285–292.
- Dodd, I.C. & Diatloff, E. (2016) Enhanced root growth of the brb (bald root barley) mutant in drying soil allows similar shoot physiological responses to soil water deficit as wild-type plants. *Functional Plant Biology*, 43, 199–206.
- Dodd, I.C., Puértolas, J., Huber, K., Pérez-Pérez, J.G., Wright, H.R. & Blackwell, M.S. (2015) The importance of soil drying and re-wetting in crop phytohormonal and nutritional responses to deficit irrigation. *Journal of Experimental Botany*, 66, 2239–52.
- Dodd, I.C., Zinovkina, N.Y., Safronova, V.I. & Belimov, A.A. (2010) Rhizobacterial mediation of plant hormone status. *Annals of Applied Biology*, 157, 361–379.
- Fahad, S., Bajwa, A.A., Nazir, U., Anjum, S.A., Farooq, A., Zohaib, A. et al. (2017) Crop production under drought and heat stress: plant responses and management options. *Frontiers in Plant Science*, 8, 1147.
- Fan, M., Bai, R., Zhao, X. & Zhang, J. (2007) Aerenchyma formed under phosphorus deficiency contributes to the reduced root hydraulic conductivity in maize roots. *Journal of Integrative Plant Biology*, 49, 598–604.
- Farooq, M., Almamari, S.A.D., Rehman, A., Al-Busaidi, W.M., Wahid, A. & Al-Ghamdi, S.S. (2021) Morphological, physiological and biochemical aspects of zinc seed priming-induced drought tolerance in faba bean. *Scientia Horticulturae*, 281, 109894.
- Felix, D.D. & Donald, A.P. (2002) Root exudates as mediators of mineral acquisition in low-nutrient environment. *Plant and Soil*, 245, 35–37.
- Fernandez, A.L., Sheaffer, C.C., Wyse, D.L., Staley, C., Gould, T.J. & Sadowsky, M.J. (2016) Associations between soil bacterial community structure and nutrient cycling functions in long-term organic farm soils following cover crop and organic fertilizer amendment. *Science of the Total Environment*, 566, 949–959.
- Galloway, A.F., Knox, P. & Krause, K. (2020) Sticky mucilages and exudates of plants: putative microenvironmental design elements with biotechnological value. *New Phytologist*, 225, 1461–1469.
- George, T.S., Brown, L.K., Ramsay, L., White, P.J., Newton, A.C., Bengough, A.G. et al. (2014) Understanding the genetic control and physiological traits associated with rhizosheath production by barley (*Hordeum vulgare*). *New Phytologist*, 203, 195–205.
- Ghezzehei, T.A. & Albalasmeh, A.A. (2015) Spatial distribution of rhizodeposits provides built-in water potential gradient in the rhizosphere. *Ecological Modelling*, 298, 53–63.
- Gruber, B.D., Giehl, R.F., Friedel, S. & Wirén, N., V.O.N. (2013) Plasticity of the *Arabidopsis* root system under nutrient deficiencies. *Plant Physiology*, 163, 161–179.
- Guenet, B., Lenhart, K., Leloup, J., Giusti-Miller, S., Pouteau, V., Mora, P. et al. (2012) The impact of long-term CO₂ enrichment and moisture levels on soil microbial community structure and enzyme activities. *Geoderma*, 170, 331–336.
- Gutschick, V.P. & Bassirirad, H. (2003) Extreme events as shaping physiology, ecology, and evolution of plants: toward a unified definition and evaluation of their consequences. *New Phytologist*, 160, 21–42.
- Haichar, F.E.Z., Roncato, M.-A. & Achouak, W. (2012) Stable isotope probing of bacterial community structure and gene expression in the rhizosphere of *Arabidopsis thaliana*. *FEMS Microbiology Ecology*, 81, 291–302.
- Haling, R.E., Brown, L.K., Bengough, A.G., Valentine, T.A., White, P.J., Young, I.M. et al. (2014) Root hair length and rhizosheath mass depend on soil porosity, strength and water content in barley genotypes. *Planta*, 239, 643–651.

- Haling, R.E., Brown, L.K., Bengough, A.G., Young, I.M., Hallett, P.D., White, P.J. et al. (2013) Root hairs improve root penetration, root-soil contact, and phosphorus acquisition in soils of different strength. *Journal of Experimental Botany*, 64, 3711–3721.
- Haling, R.E., Richardson, A.E., Culvenor, R.A., Lambers, H. & Simpson, R.J. (2010) Root morphology, root-hair development and rhizosheath formation on perennial grass seedlings is influenced by soil acidity. *Plant and Soil*, 335, 457–468.
- Haling, R.E., Yang, Z., Shadwell, N., Culvenor, R.A., Stefanski, A., Ryan, M.H. et al. (2016) Root morphological traits that determine phosphorus-acquisition efficiency and critical external phosphorus requirement in pasture species. *Functional Plant Biology*, 43, 815–826.
- Hallama, M., Pekrun, C., Mayer-Gruner, P., Uksa, M., Abdullaeva, Y., Pilz, S. et al. (2022) The role of microbes in the increase of organic phosphorus availability in the rhizosheath of cover crops. *Plant and Soil*, 1–21. <https://doi.org/10.1007/s11104-022-05340-5>
- Hallett, P.D., Marin, M., Bending, G.D., George, T.S., Collins, C.D. & Otten, W. (2022) Building soil sustainability from root–soil interface traits. *Trends in Plant Science*, 27, 688–698.
- Hanna, A.L., Youssef, H.H., Amer, W.M., Monib, M., Fayez, M. & Hegazi, N.A. (2013) Diversity of bacteria nesting the plant cover of north Sinai deserts, Egypt. *Journal of Advanced Research*, 4, 13–26.
- Hartnett, D.C., Wilson, G.W., Ott, J.P. & Setshogo, M. (2013) Variation in root system traits among African semi-arid savanna grasses: implications for drought tolerance. *Australian Ecology*, 38, 383–392.
- Hidayati, N. & Anas, I. (2016) Photosynthesis and transpiration rates of rice cultivated under the system of rice intensification and the effects on growth and yield. *HAYATI Journal of Biosciences*, 23, 67–72.
- Hinsinger, P. (2001) Bioavailability of soil inorganic P in the rhizosphere as affected by root-induced chemical changes: a review. *Plant and Soil*, 237, 173–195.
- Hocking, P., Keerthisinghe, G., Smith, F. & Randall, P. (1997) Comparison of the ability of different crop species to access poorly-available soil phosphorus. *Plant nutrition for sustainable food production and environment*. Springer.
- Holz, M., Zarebanadkouki, M., Kuzyakov, Y., Pausch, J. & Carminati, A. (2017) Root hairs increase rhizosphere extension and carbon input to soil. *Annals of Botany*, 121, 61–69.
- Hu, X., Roberts, D.P., Xie, L., Maul, J.E., Yu, C., Li, Y. et al. (2013) Development of a biologically based fertilizer, incorporating *Bacillus megaterium* A6, for improved phosphorus nutrition of oilseed rape. *Canadian Journal of Microbiology*, 59, 231–236.
- Hunter, P.J., Teakle, G.R. & Bending, G.D. (2014) Root traits and microbial community interactions in relation to phosphorus availability and acquisition, with particular reference to Brassica. *Frontiers in Plant Science*, 5, 27.
- Jacoby, R., Peukert, M., Succurro, A., Koprivova, A. & Kopriva, S. (2017) The role of soil microorganisms in plant mineral nutrition—current knowledge and future directions. *Frontiers in Plant Science*, 8, 1617.
- James, R.A., Weligama, C., Verbyla, K., Ryan, P.R., Rebetzke, G.J., Rattey, A. et al. (2016) Rhizosheaths on wheat grown in acid soils: phosphorus acquisition efficiency and genetic control. *Journal of Experimental Botany*, 67, 3709–3718.
- Jeong, S., Moon, H.S., Shin, D. & Nam, K. (2013) Survival of introduced phosphate-solubilizing bacteria (PSB) and their impact on microbial community structure during the phytoextraction of Cd-contaminated soil. *Journal of Hazardous Materials*, 263, 441–449.
- Jin, J., Wang, G., Liu, X., Pan, X., Herbert, S.J. & Tang, C. (2006) Interaction between phosphorus nutrition and drought on grain yield, and assimilation of phosphorus and nitrogen in two soybean cultivars differing in protein concentration in grains. *Journal of Plant Nutrition*, 29, 1433–1449.
- Karanja, J.K., Aslam, M.M., Qian, Z., Yankey, R., Dodd, I.C. & Weifeng, X. (2021) Absciscic acid mediates drought-enhanced rhizosheath formation in tomato. *Frontiers in Plant Science*, 12, 658787.
- Kawasaki, A., Donn, S., Ryan, P.R., Mathesius, U., Devilla, R., Jones, A. et al. (2016) Microbiome and exudates of the root and rhizosphere of *Brachypodium distachyon*, a model for wheat. *PLoS One*, 11, e0164533.
- Khan, M.S., Zaidi, A. & Ahmad, E. (2014) Mechanism of Phosphate Solubilization and Physiological Functions of Phosphate-Solubilizing Microorganisms. In: Khan, M.S., Zaidi, A. & Musarrat, J. (Eds.) *Phosphate Solubilizing Microorganisms: Principles and Application of Microphos Technology*. Springer International Publishing.
- Khan, N. & Bano, A. (2019) Exopolysaccharide producing rhizobacteria and their impact on growth and drought tolerance of wheat grown under rainfed conditions. *PLoS One*, 14, e0222302.
- Kidd, D.R., Ryan, M.H., Haling, R.E., Lambers, H., Sandral, G.A., Yang, Z. et al. (2016) Rhizosphere carboxylates and morphological root traits affects water flow in soils. *Plant and Soil*, 402, 77–89.
- Koebernick, N., Schlüter, S., Blaser, S.R.G.A. & Vetterlein, D. (2018) Root-soil contact dynamics of *Vicia faba* in sand. *Plant and Soil*, 431, 417–431.
- Krishnapriya, V. & Pandey, R. (2016) Root exudation index: screening organic acid exudation and phosphorus acquisition efficiency in soybean genotypes. *Crop and Pasture Science*, 67, 1096–1109.
- Kroener, E., Zarebanadkouki, M., Kaestner, A. & Carminati, A. (2014) Nonequilibrium water dynamics in the rhizosphere: how mucilage affects water flow in soils. *Water Resources Research*, 50, 6479–6495.
- Kuzyakov, Y. & Domanski, G. (2000) Carbon input by plants into the soil. Review. *Journal of Plant Nutrition and Soil Science*, 163, 421–431.
- Lambers, H., Clode, P.L., Hawkins, H.J., Laliberté, E., Oliveira, R.S., Reddell, P. et al. (2018) Metabolic adaptations of the non-mycotrophic Proteaceae to soils with low phosphorus availability. *Annual Plant Reviews Online*, 48, 289–335.
- Lambers, H., Martinoia, E. & Renton, M. (2015) Plant adaptations to severely phosphorus-impooverished soils. *Current Opinion in Plant Biology*, 25, 23–31.
- Li, J., Li, Y., Yin, Z., Jiang, J., Zhang, M., Guo, X. et al. (2017) Os ASR 5 enhances drought tolerance through a stomatal closure pathway associated with ABA and H₂O₂ signalling in rice. *Plant Biotechnology Journal*, 15, 183–196.
- Liang, C., Piñeros, M.A., Tian, J., Yao, Z., Sun, L., Liu, J. et al. (2013) Low pH, aluminum, and phosphorus coordinately regulate malate exudation through GmALMT1 to improve soybean adaptation to acid soils. *Plant Physiology*, 161, 1347–1361.
- Lichner, L., Babejová, N. & Dekker, L.W. (2002) Effects of kaolinite and drying temperature on the persistence of soil water repellency induced by humic acids. *Rostlinna Vyroba*, 48, 203–207.
- Liebersbach, H., Steingrobe, B. & Claassen, N. (2004) Roots regulate ion transport in the rhizosphere to counteract reduced mobility in dry soil. *Plant and Soil*, 260, 79–88.
- Liu, T.-Y., Chen, M.-X., Zhang, Y., Zhu, F.-Y., Liu, Y.-G., Tian, Y. et al. (2019) Comparative metabolite profiling of two switchgrass ecotypes reveals differences in drought stress responses and rhizosheath weight. *Planta*, 250, 1355–1369.
- Liu, T.-Y., Ye, N., Song, T., Cao, Y., Gao, B., Zhang, D. et al. (2019) Rhizosheath formation and involvement in foxtail millet (*Setaria italica*) root growth under drought stress. *Journal of Integrative Plant Biology*, 61, 449–462.
- Lynch, J.P. (2019) Root phenotypes for improved nutrient capture: an underexploited opportunity for global agriculture. *New Phytologist*, 223, 548–564.
- Lynch, J.P. & Brown, K.M. (2001) Topsoil foraging—an architectural adaptation of plants to low phosphorus availability. *Plant and Soil*, 237, 225–237.

- Ma, J.F., Ryan, P.R. & Delhaize, E. (2001) Aluminium tolerance in plants and the complexing role of organic acids. *Trends in Plant Science*, 6, 273–278.
- Ma, Q., Rengel, Z. & Rose, T. (2009) The effectiveness of deep placement of fertilisers is determined by crop species and edaphic conditions in Mediterranean-type environments: a review. *Soil Research*, 47, 19–32.
- Macias-Benitez, S., Garcia-Martinez, A.M., Caballero Jimenez, P., Gonzalez, J.M., Tejada Moral, M. & Parrado Rubio, J. (2020) Rhizospheric organic acids as biostimulants: monitoring feedbacks on soil microorganisms and biochemical properties. *Frontiers in Plant Science*, 11, 633.
- Mahmood, T., Mehnaz, S., Fleischmann, F., Ali, R., Hashmi, Z.H. & Iqbal, Z. (2014) Soil sterilization effects on root growth and formation of rhizosheaths in wheat seedlings. *Pedobiologia*, 57, 123–130.
- Marasco, R., Mosqueira, M.J., Fusi, M., Ramond, J.-B., Merlino, G., Booth, J.M. et al. (2018) Rhizosphere microbial community assembly of sympatric desert spargrasses is independent of the plant host. *Microbiome*, 6, 1–18.
- Marin, M., Feeney, D.S., Brown, L.K., Naveed, M., Ruiz, S., Koebernick, N. et al. (2021) Significance of root hairs for plant performance under contrasting field conditions and water deficit. *Annals of Botany*, 128, 1–16.
- Marr, E. (2021) *Unlocking the genetic potential of the root system, rhizosphere mucilage and microbiome of wheat (Triticum aestivum L.)*.
- Mathesius, U. (2015) Soil: root interface. In: Munns, R., Schmidt, S., Beveridge, C. (Eds.) *Plants in action*, 2nd ed. Australia and New Zealand: Australian Society of Plant Scientists and New Zealand Society of Plant Biologists.
- Mccully, M.E. & Canny, M. (1988) Pathways and processes of water and nutrient movement in roots. *Plant and Soil*, 111, 159–170.
- Mccully, M.E. (1999) Roots in soil: unearthing the complexities of roots and their rhizospheres. *Annual Review of Plant Physiology and Plant Molecular Biology*, 50, 695–718.
- Moradi, A.B., Carminati, A., Lamparter, A., Woche, S.K., Bachmann, J., Vetterlein, D. et al. (2012) Is the rhizosphere temporarily water repellent? *Vadose Zone Journal*, 11, vzj2011–0120.
- Morel, J.L., Habib, L., Plantureux, S. & Guckert, A. (1991) Influence of maize root mucilage on soil aggregate stability. *Plant and Soil*, 136, 111–119.
- Morison, J.I. & Morecroft, M.D. 2006. *Plant growth and climate change*.
- Naveed, M., Brown, L., Raffan, A., George, T.S., Bengough, A.G., Roose, T. et al. (2017) Plant exudates may stabilize or weaken soil depending on species, origin and time. *European Journal of Soil Science*, 68, 806–816.
- Nazari, M. (2021) Plant mucilage components and their functions in the rhizosphere. *Rhizosphere*, 18, 100344.
- Ndour, P.M.S., Heulin, T., Achouak, W., Laplace, L. & Cournac, L. (2020) The rhizosphere: from desert plants adaptation to crop breeding. *Plant and Soil*, 456, 1–13.
- Neilson, J.W., Califf, K., Cardona, C., Copeland, A., Treuren, V.A.N., Josephson, W. et al. (2017) Significant impacts of increasing aridity on the arid soil microbiome. *mSystems*, 2, e00195–16.
- North, G.B. & Nobel, P.S. (1997) Drought-induced changes in soil contact and hydraulic conductivity for roots of *Opuntia ficus-indica* with and without rhizosheaths. *Plant and Soil*, 191, 249–258.
- Norton, G.J., Shafaei, M., Travis, A.J., Deacon, C.M., Danku, J., Pond, D. et al. (2017) Impact of alternate wetting and drying on rice physiology, grain production, and grain quality. *Field Crops Research*, 205, 1–13.
- Othman, A.A., Amer, W.M., Fayed, M. & Hegazi, N. (2004) Rhizosphere of Sinai desert plants is a potential repository for associative diazotrophs. *Microbiological Research*, 159, 285–293.
- Pan, Y., Seymour, G.B., Lu, C., Hu, Z., Chen, X. & Chen, G. (2012) An ethylene response factor (ERF5) promoting adaptation to drought and salt tolerance in tomato. *Plant Cell Reports*, 31, 349–60.
- Pang, J., Ryan, M.H., Siddique, K.H. & Simpson, R.J. (2017) Unwrapping the rhizosphere. *Plant and Soil*, 418, 129–139.
- Pascale, A., Proietti, S., Pantelides, I.S. & Stringlis, I.A. (2020) Modulation of the root microbiome by plant molecules: the basis for targeted disease suppression and plant growth promotion. *Frontiers in Plant Science*, 10, 1741.
- Passioura, J.B. (1988) Response to Dr P. J. Kramer's article 'Changing concepts regarding plant water relations'. *Plant, Cell & Environment*, 11, 569–571.
- Phillips, R.P. (2007) Towards a rhizo-centric view of plant-microbial feedbacks under elevated atmospheric CO₂. *New Phytologist*, 173, 664–667.
- Phillips, R.P., Ehlitz, Y., Bier, R. & Bernhardt, E.S. (2008) New approach for capturing soluble root exudates in forest soils. *Functional Ecology*, 22, 990–999.
- Pii, Y., Marastoni, L., Springeth, C., Fontanella, M.C., Beone, G.M., Cesco, S. et al. (2016) Modulation of Fe acquisition process by *Azospirillum brasilense* in cucumber plants. *Environmental and Experimental Botany*, 130, 216–225.
- Pii, Y., Penn, A., Terzano, R., Crecchio, C., Mimmo, T. & Cesco, S. (2015) Plant-microorganism-soil interactions influence the Fe availability in the rhizosphere of cucumber plants. *Plant Physiology and Biochemistry*, 87, 45–52.
- Pires, L.F., Bacchi, O.O.S. & Reichardt, K. (2007) Assessment of soil structure repair due to wetting and drying cycles through 2D tomographic image analysis. *Soil and Tillage Research*, 94, 537–545.
- Potopová, V., Boroneanț, C., Boincean, B. & Soukup, J. (2016) Impact of agricultural drought on main crop yields in the Republic of Moldova. *International Journal of Climatology*, 36, 2063–2082.
- Qureshi, M.A., Ahmad, Z.A., Akhtar, N., Iqbal, A., Mujeed, F. & Shakir, M.A. (2012) Role of phosphate solubilizing bacteria (PSB) in enhancing P availability and promoting cotton growth. *Journal of Animal and Plant Sciences*, 22, 204–210.
- Rabbi, S.M.F., Tighe, M.K., Flavel, R.J., Kaiser, B.N., Guppy, C.N., Zhang, X. et al. (2018) Plant roots redesign the rhizosphere to alter the three-dimensional physical architecture and water dynamics. *New Phytologist*, 219, 542–550.
- Radin, J.W. & Eidenbock, M.P. (1984) Hydraulic conductance as a factor limiting leaf expansion of Phosphorus-Deficient cotton plants. *Plant Physiology*, 75, 372–377.
- Raghothama, K. (1999) Phosphate acquisition. *Annual review of plant biology*, 50, 665–693.
- Read, D.B. & Gregory, P.J. (1997) Surface tension and viscosity of axenic maize and lupin root mucilages. *New Phytologist*, 137, 623–628.
- Rejesus, R.M., Palis, F.G., Rodriguez, D.G.P., Lampayan, R.M. & Bouman, B.A.M. (2011) Impact of the alternate wetting and drying (AWD) water-saving irrigation technique: evidence from rice producers in the Philippines. *Food Policy*, 36, 280–288.
- Richardson, A.E. & Simpson, R.J. (2011) Soil microorganisms mediating phosphorus availability update on microbial phosphorus. *Plant Physiology*, 156, 989–996.
- Roberson Emily, B. & Firestone Mary, K. (1992) Relationship between desiccation and exopolysaccharide production in a soil *Pseudomonas* sp. *Applied and Environmental Microbiology*, 58, 1284–1291.
- Rodriguez, H., Gonzalez, T., Goire, I. & Bashan, Y. (2004) Gluconic acid production and phosphate solubilization by the plant growth-promoting bacterium *Azospirillum* spp. *Naturwissenschaften*, 91, 552–555.
- Rosenzweig, R., Shavit, U. & Furman, A. (2009) The influence of biofilm spatial distribution scenarios on hydraulic conductivity of unsaturated soils. *Vadose Zone Journal*, 8, 1080–1084.
- Rouphael, Y., Cardarelli, M., Schwarz, D., Franken, P. & Colla, G. (2012) Effects of drought on nutrient uptake and assimilation in vegetable crops, *Plant responses to drought stress*. Springer.
- Sanaullah, M., Blagodatskaya, E., Chabbi, A., Rumpel, C. & Kuzyakov, Y. (2011) Drought effects on microbial biomass and enzyme activities

- in the rhizosphere of grasses depend on plant community composition. *Applied Soil Ecology*, 48, 38–44.
- Sandhya, V., Sk, Z.A., Grover, M., Reddy, G. & Venkateswarlu, B. (2009) Alleviation of drought stress effects in sunflower seedlings by the exopolysaccharides producing *Pseudomonas putida* strain GAP-P45. *Biology and Fertility of Soils*, 46, 17–26.
- Sardans, J. & Peñuelas, J. (2012) The role of plants in the effects of global change on nutrient availability and stoichiometry in the plant-soil system. *Plant Physiology*, 160, 1741–1761.
- Sawada, K., Funakawa, S. & Kosaki, T. (2016) Short-term respiration responses to drying–rewetting in soils from different climatic and land use conditions. *Applied Soil Ecology*, 103, 13–21.
- Scagliola, M., Pii, Y., Mimmo, T., Cesco, S., Ricciuti, P. & Crecchio, C. (2016) Characterization of plant growth promoting traits of bacterial isolates from the rhizosphere of barley (*Hordeum vulgare* L.) and tomato (*Solanum lycopersicon* L.) grown under Fe sufficiency and deficiency. *Plant Physiology and Biochemistry*, 107, 187–196.
- Segura, A. & Ramos, J.L. (2013) Plant-bacteria interactions in the removal of pollutants. *Current Opinion in Biotechnology*, 24, 467–473.
- Siopongco, J.D.L.C., Sekiya, K., Yamauchi, A., Egdane, J., Ismail, A.M. & Wade, L.J. (2008) Stomatal responses in rainfed lowland rice to partial soil drying; evidence for root signals. *Plant Production Science*, 11, 28–41.
- Star, L., Matan, O., Dardanelli, M.S., Kapulnik, Y., Burdman, S. & Okon, Y. (2012) The *Vicia sativa* spp. nigra-Rhizobium leguminosarum bv. viciae symbiotic interaction is improved by *Azospirillum brasilense*. *Plant and Soil*, 356, 165–174.
- Suzuki, N., Rivero, R.M., Shulaev, V., Blumwald, E. & Mittler, R. (2014) Abiotic and biotic stress combinations. *New Phytologist*, 203, 32–43.
- Tahir, M., Mirza, M.S., Hameed, S., Dimitrov, M.R. & Smidt, H. (2015) Cultivation-based and molecular assessment of bacterial diversity in the rhizosheath of wheat under different crop rotations. *PLoS One*, 10, e0130030.
- Tariq, A., Pan, K., Olatunji, O.A., Graciano, C., Li, Z. & Sun, F. et al. (2017) Phosphorus application improves drought tolerance of phoebe zhennan. *Frontiers in Plant Science*, 8, 1561.
- Terrazas, R.A., Giles, C., Paterson, E., Robertson-Albertyn, S., Cesco, S. & Mimmo, T. et al. (2016) Plant-microbiota interactions as a driver of the mineral turnover in the rhizosphere. *Advances in Applied Microbiology*. Elsevier.
- Tesfaye, M., Temple, S.J., Allan, D.L., Vance, C.P. & Samac, D.A. (2001) Overexpression of malate dehydrogenase in transgenic alfalfa enhances organic acid synthesis and confers tolerance to aluminum. *Plant Physiology*, 127, 1836–1844.
- Ticconi, C.A., Delatorre, C.A., Lahner, B., Salt, D.E. & Abel, S. (2004) Arabidopsis pdr2 reveals a phosphate-sensitive checkpoint in root development. *The Plant Journal*, 37, 801–814.
- Turan, M., Gulluce, M., Von Wirén, N. & Sahin, F. (2012) Yield promotion and phosphorus solubilization by plant growth-promoting rhizobacteria in extensive wheat production in Turkey. *Journal of Plant Nutrition and Soil Science*, 175, 818–826.
- Turner, B.L., Driessen, J.P., Haygarth, P.M. & Mckelvie, I.D. (2003) Potential contribution of lysed bacterial cells to phosphorus solubilisation in two rewetted Australian pasture soils. *Soil Biology and Biochemistry*, 35, 187–189.
- Vissenberg, K., Claeijs, N., Balcerowicz, D. & Schoenaers, S. (2020) Hormonal regulation of root hair growth and responses to the environment in Arabidopsis. *Journal of Experimental Botany*, 71, 2412–2427.
- Walker, T.S., Bais, H.P., Grotewold, E. & Vivanco, J.M. (2003) Root exudation and rhizosphere biology. *Plant Physiology*, 132, 44–51.
- Wang, L., Lu, S., Zhang, Y., Li, Z., Du, X. & Liu, D. (2014) Comparative genetic analysis of Arabidopsis purple acid phosphatases AtPAP10, AtPAP12, and AtPAP26 provides new insights into their roles in plant adaptation to phosphate deprivation. *Journal of Integrative Plant Biology*, 56, 299–314.
- Wang, X., Du, T., Huang, J., Peng, S. & Xiong, D. (2018) Leaf hydraulic vulnerability triggers the decline in stomatal and mesophyll conductance during drought in rice. *Journal of Experimental Botany*, 69, 4033–4045.
- Watanabe, T., Katayanagi, N., Agbisit, R., Llorca, L., Hosen, Y. & Asakawa, S. (2021) Influence of alternate wetting and drying water-saving irrigation practice on the dynamics of Gallionella-related iron-oxidizing bacterial community in paddy field soil. *Soil Biology and Biochemistry*, 152, 108064.
- Watt, M., Mccully, M. & Jeffree, C. (1993) Plant and bacterial mucilages of the maize rhizosphere: comparison of their soil binding properties and histochemistry in a model system. *Plant and Soil*, 151, 151–165.
- Watt, M., Mccully, M.E. & Canny, M.J. (1994) Formation and stabilization of rhizosheaths of *Zea mays* L. (Effect of soil water content). *Plant Physiology*, 106, 179–186.
- Wei, M.A., Xue-Xian, L.T. and Chun-Jian, L.T. (2011) Modulation of soil particle size and nutrient availability in the maizerhizosheath. *Pedosphere*, 21(4), 483–490.
- White, R.G. & Kirkegaard, J.A. (2010) The distribution and abundance of wheat roots in a dense, structured subsoil—implications for water uptake. *Plant, Cell & Environment*, 33, 133–148.
- Xia, Z., He, Y., Zhou, B., Korpelainen, H. & Li, C. (2020) Sex-related responses in rhizosphere processes of dioecious *Populus cathayana* exposed to drought and low phosphorus stress. *Environmental and Experimental Botany*, 175, 104049.
- Xu, F., Song, T., Wang, K., Xu, W., Chen, G., Xu, M. et al. (2020) Frequent alternate wetting and drying irrigation mitigates the effect of low phosphorus on rice grain yield in a 4-year field trial by increasing soil phosphorus release and rice root growth. *Food and Energy Security*, 9, e206.
- Yi, K., Menand, B., Bell, E. & Dolan, L. (2010) A basic helix-loop-helix transcription factor controls cell growth and size in root hairs. *Nature Genetics*, 42, 264–7.
- Yin, Y. & Wang, J. (2016) Changes in microbial community during biohydrogen production using gamma irradiated sludge as inoculum. *Bioresource Technology*, 200, 217–222.
- Young, I.M. (1995) Variation in moisture contents between bulk soil and the rhizosheath of wheat (*Triticum aestivum* L. cv. Wembley). *New Phytologist*, 130, 135–139.
- Yuan, J., Zhao, M., Li, R., Huang, Q., Raza, W., Rensing, C. et al. (2017) Microbial volatile compounds alter the soil microbial community. *Environmental Science and Pollution Research*, 24, 22485–22493.
- Zahoor, A., Waraich, E.A., Rehman, M.Z.U., Ayub, M.A., Usman, M., Alharby, H. et al. (2021) Foliar application of phosphorus enhances photosynthesis and biochemical characteristics of maize under drought stress. *Phyton*, 90, 503–514.
- Zarebanadkouki, M. & Carminati, A. (2014) Reduced root water uptake after drying and rewetting. *Journal of Plant Nutrition and Soil Science*, 177, 227–236.
- Zeppenfeld, T., Balkenhol, N., Kóvacs, K. & Carminati, A. (2017) Rhizosphere hydrophobicity: a positive trait in the competition for water. *PLoS One*, 12, e0182188.
- Zhang, H., Shi, L., Lu, H., Shao, Y., Liu, S. & Fu, S. (2020) Drought promotes soil phosphorus transformation and reduces phosphorus bioavailability in a temperate forest. *Science of the Total Environment*, 732, 139295.
- Zhang, Y., Du, H., Gui, Y., Xu, F., Liu, J., Zhang, J. et al. (2020) Moderate water stress in rice induces rhizosheath formation associated with abscisic acid and auxin responses. *Journal of Experimental Botany*, 71, 2740–2751.
- Zhang, Y., Du, H., Xu, F., Ding, Y., Gui, Y., Zhang, J. et al. (2020) Root-bacteria associations boost rhizosheath formation in moderately dry soil through ethylene responses. *Plant Physiology*, 183, 780–792.
- Zhao, S., Li, K., Zhou, W., Qiu, S., Huang, S. & He, P. (2016) Changes in soil microbial community, enzyme activities and organic matter fractions under long-term straw return in north-central China. *Agriculture, Ecosystems & Environment*, 216, 82–88.

- Zheng, Y., Hivrale, V., Zhang, X., Valliyodan, B., Lelandais-Brière, C., Farmer, A.D. et al. (2016) Small RNA profiles in soybean primary root tips under water deficit. *BMC Systems Biology*, 10, 126.
- Zickenrott, I.M., Woche, S.K., Bachmann, J., Ahmed, M.A. & Vetterlein, D. (2016) An efficient method for the collection of root mucilage from different plant species—a case study on the effect of mucilage on soil water repellency. *Journal of Plant Nutrition and Soil Science*, 179, 294–302.

How to cite this article: Aslam, M.M., Karanja, J.K., Dodd, I.C., Waseem, M. & Weifeng, X. (2022) Rhizosheath: An adaptive root trait to improve plant tolerance to phosphorus and water deficits? *Plant, Cell & Environment*, 45, 2861–2874.

<https://doi.org/10.1111/pce.14395>